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Title: How's the weather in there? Hurricanes drive short-term declines in the abundance of direct life cycle parasites of riverine fish

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Key policy highlights

- We investigated how parasites of riverine fish are affected by hurricane events relative to control conditions using museum collections of fish collected from the Pearl River between 1963 and 2005.
- We found that hurricanes drive short-term declines in the abundance of direct life cycle parasites on short time scales, but this effect dissipates at longer (6-month) time scales.
- Complex life cycle parasites were not affected by hurricanes.
- The increasing frequency of hurricanes might pose a threat to populations of direct life cycle parasites.

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Abstract

Hurricanes have increased in frequency and magnitude due to anthropogenic climate change. These events can cause outbreaks of parasitic diseases in humans, but little is known about how they impact parasite abundances for wildlife hosts. The southeastern United States Gulf region has been affected by hundreds of hurricanes in the past 50 years, with frequency and intensity increasing through time. Prior studies have quantified the response of wildlife parasites to individual storms, but have never systematically evaluated whether these effects are consistent across storms or different from control conditions. From the Royal D. Suttkus Fish Collection at the Tulane University Biodiversity Research Institute, we selected and performed parasitological dissections of 1,197 preserved fish across 7 fish species collected between 1963 and 2005 from the Pearl River in Louisiana. We evaluated the effect of hurricanes on parasite abundance using generalized linear mixed models that contrasted responses of fish parasites to 24 hurricanes against control conditions, finding that change in parasite abundance in response to a hurricane was dependent upon parasite taxonomic group and transmission strategy, with direct life cycle parasites (copepods and monogenes) negatively impacted by hurricanes on short time scales, while complex life cycle parasites (acanthocephalan, cestode, nematode, trematode adult, and trematode metacercaria) were not affected by hurricanes. This investigation of hurricane impacts on parasitism in wildlife populations across a 42-year period identifies parasite traits that modulate resistance to storm conditions. Continued anthropogenic climate change will cause more frequent and intense hurricane events, which our research suggests could lead to decreased abundances of direct life cycle parasites, affecting riverine fish parasite community composition and, possibly, ecosystem function. Such declines of parasites may signal broader ecosystem stress under increasing hurricane frequency, with cascading effects on their fish hosts and other species that depend on the complex ecological interactions connecting them.

Introduction

Anthropogenic climate change is increasing the frequency and intensity of extreme weather events like hurricanes (Michener et al. 1997), with the frequency of category four and five hurricanes increasing by 30% with every 1°C increase in temperature (Holland and Bruyère 2014). Hurricanes can severely alter the physical structure of freshwater habitats, causing runoff pollution, salinity spikes, deposition and suspension of debris, and direct destruction of structures including mangrove forests (Mallin and Corbett 2006; Patrick et al. 2020). Furthermore, storm surge from hurricanes can both force seawater upriver and deposit large amounts of marsh vegetation in lower reaches of rivers (Cahoon 2006). Aquatic plant and animal populations are particularly impacted by floods, which can temporarily inundate a river or body of water with excess nutrients and create hypoxic environments detrimental to benthic species (Patrick et al. 2020). High winds and rain can suspend sediment in the water column of rivers and lakes, obscuring light and leading to decreases in biomass of primary producers like phytoplankton (Havens et al. 2011). These conditions also lead to decreases in predator-prey interactions, the density of fishes, reductions in vegetative cover, and an increase in the abundance of macro-zooplankton (Havens et al. 2011).

The environmental effects of hurricanes can affect transmission dynamics of infectious diseases and parasites. Infectious disease transmission in humans has been observed to increase after hurricane events by contaminating water with disease agents (e.g., *Leptospira* bacteria for leptospirosis and *Vibrio cholerae* for cholera) (Nelson et al. 2009; Rebaudet et al. 2013; Al-orry et al. 2016; Barragan et al. 2017). Hurricanes can force humans into closer contact with each other, furthering the spread of directly transmitted infectious diseases which can be compounded by storm-induced disruptions to freshwater sources (Roman et al. 1994; Howard et al. 1996; Mallin and Corbett 2006; Murray et al. 2012; McMichael 2015). However, we know little about how hurricanes affect infectious disease dynamics in wildlife populations (Cely 1991; Levy et al. 2011; Maness 2019; Filipe et al. 2020). Aguirre-Macedo et al. (2011) found that, following a hurricane, *Cerithidea pliculosa* snails formerly abundant in a coastal lagoon of the Yucatan Peninsula were absent for 6 months. These snails are important intermediate hosts for fish-borne trematode parasites, which were themselves absent from this ecosystem for 14 months following the hurricane event (Aguirre-Macedo et al. 2011). Snail biomass was significantly lower following the hurricane, and researchers estimated that it would take between six and ten years for snail biomass to recover to 90% of before-hurricane values and that trematode infection prevalence would take between six and eleven years to recover to 90% of before-hurricane values (Aguirre-Macedo et al. 2011). Effects of hurricanes have been reported on direct life cycle protozoan parasites (specifically *Balantidium coli*) in Rhesus macaques of Puerto Rico: *B. coli* infection intensity was lower following a hurricane event than before the event (Pavez-Fox et al. 2024). *Balantidium coli* cysts thrive in high humidity environments protected from direct sunlight, conditions that became less common after Hurricane María, which decimated the island's vegetation cover (Schuster and Ramirez-Avila 2008; Pavez-Fox et al. 2024). For the same population of Rhesus macaques, the prevalence of two nematode species, *Strongyloides fulleborni* (complex life cycle) and *Trichuris trichura* (direct life cycle), increased following a storm, suggesting that after-storm crowding may facilitate the spread of parasites, a trend similar to those observed in human populations following hurricane events (Behie et al. 2014; Pavez-Fox et al. 2024). These examples illustrate that hurricanes impact parasite populations in wildlife, especially for species that reside in or around aquatic environments (Aguirre-Macedo et al. 2011; Pavez-Fox et al. 2024). But, importantly, these studies have all focused on changes manifested in response to single hurricane events. To our knowledge, no study has synthesized data across hurricanes, or even compared hurricane-affected

areas to unaffected controls; therefore, a gap remains in our understanding of how wildlife parasites respond to the disturbance caused by hurricanes.

Prior studies have noted the importance of transmission strategy in modulating the response of parasites to hurricane events. Parasites can be categorized into two broad transmission strategies: complex life cycle parasites rely upon two or more host species to complete their life cycle, while direct life cycle parasites can be transmitted between conspecifics (Olsen 1986). Complex life cycle parasites are expected to be more prone to negative impacts from hurricanes. First, if any of the hosts involved in the life cycle are locally removed following a hurricane, the parasite will not be able to persist in that particular environment. Moreover, free-living infective larvae are known to be sensitive to changes in abiotic conditions, potentially decreasing in lifespan after exposure to pollutants and salinity spikes though relationships are species-specific (Cross et al. 2001; Koprivnikar et al. 2010; Prasopdee et al. 2020; Jones et al. 2020). However, other outcomes are possible: complex life cycle parasites may be at an advantage during a short-term disturbance like a hurricane, because, in a manner of speaking, their eggs are distributed across many baskets: different infectious stages of a complex life cycle parasite infect different hosts, so that if any one host is temporarily wiped out by hurricane conditions, there is a reservoir of infectious propagules in the upstream host ready to continue the life cycle once the extinguished host recovers. These parasites could therefore maintain their populations by sheltering in unaffected hosts, allowing them to ride out the storm and resist any short term change in the system (Molnár et al. 2013; Aleuy and Kutz 2020). Direct life cycle parasites might have only one host species, and as such could be protected from hurricane events if their host is unaffected. Conversely, a direct life cycle parasite might not have any reservoir or back-up hosts in which they could ride out a storm, so if their host is affected, the parasite will be affected as well. Disruptions to any host organism can manifest in spatio-temporal mismatches, wherein a misalignment of hosts and parasites encountering each other in space (e.g., range shifts, shifts in habitat use) or time (e.g., disruptions of normal seasonal patterns) detrimentally affects parasite development and reproduction (Saino et al. 2009; Paull and Johnson 2014; Poulin and Lagrue 2015; McDevitt-Galles et al. 2020). When sustained in an ecosystem for a long period of time, such mismatches can lead to a decrease in that parasite's abundance (O'Donnell et al. 2011).

We set out to quantify the effects of hurricanes on parasite populations in wildlife. Our goal was to understand how these storms affect the abundances of parasites and whether transmission strategy (direct or complex) modulates that response. We hypothesized that complex life cycle parasites would be negatively affected, as they require more host species and more steps to successfully reproduce, meaning disruption at any stage could lead to declines in abundance (Rauch et al. 2005; Kutz et al. 2009; Poulin and Lagrue 2015; Aleuy and Kutz 2020). We also hypothesized that direct life cycle parasites would be less affected by storms, since they do not require more than one host species and may be more flexible as host community composition shifts post-hurricane (Anderson 1980; Niewiadomska and Pojmańska 2011; Aleuy and Kutz 2020). To test these hypotheses, we parasitologically examined riverine fish collected from a hurricane-prone area over a 42-year period that encompassed 25 hurricane events. These data allow us to quantify the effects of hurricane events on parasites of riverine fish and investigate whether transmission strategy, location of infection, or parasite taxon modulate parasite responses to hurricanes.

Methods

Specimens

This study used the Royal D. Suttkus Fish Collection housed at the Tulane University Biodiversity Research Institute (TUBRI) in Belle Chasse, Louisiana, USA – a collection that includes numerous specimens collected from the Pearl River in Louisiana between 1963 and 2005 (Tulane University Biodiversity Research Institute)(Figure 1). Given the robustness of this collection, our team identified seven species of fishes that were sufficiently abundant across both the span of the collection locations and time (*Gambusia affinis*, *Carpiodes velifer*, *Hybognathus nuchalis*, *Ictalurus punctatus*, *Notropis atherinoides*, *Percina vigil*, and *Pimephales vigilax*). We selected fish with sufficient within-lot replication (more than 4 fish individuals per jar) and instituted an upper and lower body size limit to keep body size constant through time, as we noticed a decrease in body size through time when fish were randomly selected (> 27 mm for *Gambusia affinis*; > 30 mm for *Carpiodes velifer*, *Percina vigil*, and *Pimephales vigilax*; > 35 mm for *Hybognathus nuchalis* and *Notropis atherinoides*; and between 35 mm and 150 mm for *Ictalurus punctatus* - all fish except *Ictalurus punctatus* had no upper limit) (Supplemental Figure 9). We used stratified random sampling to select final lots from the total collection based on decade and location of collection. For example, we selected the same number of lots from the decade 1980-1989 as we did from the decade 1990-1999 for each of the two main sampling location criteria (see experimental design below), aiming for five lots per decade-location combination (Diaz-Morales et al. 2024).

Data collection

We employed a semi-destructive method of dissection that would allow us to accurately inspect for parasites but preserve the host specimen for retention in the Fish Collection (Wood et al. 2023a). We focused on the right side of each fish, collecting the right eye, pectoral and pelvic fins, and gills. We made an incision from just below the gill isthmus to the vent and removed all visceral organs. The organs and tissues were placed into a container with 70% ethanol and examined for parasites. Parasite counts from bilateral organs (i.e., eye) were doubled. The skin, caudal fin, anal fin, dorsal fin, buccal cavity, and body cavity were examined intact. Once all parasites had been removed and vouchered in 70% ethanol, the fish was placed in a container with its tissues and organs, labelled, and returned to the museum collection jar (Wood, et al. 2023a). Parasites were identified to the lowest possible taxonomic level and were sorted based on transmission strategy and higher taxonomic category for analysis as follows: direct (i.e., monogenean, copepod); complex (i.e., acanthocephalan, cestode, trematode adult, trematode metacercaria, myxozoan, and nematode). In the case of myxozoans, species producing pseudocysts and species infecting the gall bladder of *C. velifer*, *G. affinis*, and *P. vigil* were assessed. For pseudocyst-producing myxozoans, abundance was defined as the number of pseudocysts (containing hundreds of spores) per fish. Myxozoans infecting the gall bladder were recorded as present or absent.

Experimental design

This study was part of a larger project concerned with estimating the impacts of industrial pollution on parasite burdens of riverine fishes. As such, locations above (control) and below (impact) a pulp mill were selected to test the effects of pollution on parasite communities through time. Pollution was not a central aspect of this sub-project. Therefore, we account in our models for the sampling location via inclusion of a 'mill_location' fixed effect, but do not explore the effects of pollution on parasite populations further.

Unlike the larger, pollution-focused project, the project described herein was designed to assess the effects of hurricanes on parasites in riverine fish. To that end, we used a Before-After-Control-Impact (BACI) design, where the ‘impact’ is hurricane presence; the two main effects (before-after, control-impact) interact to create four categories (before hurricane, after hurricane, before control, after control). In order to split our data into BACI categories, we identified all hurricane events that occurred between the collection date of our first (1963) and last (2005) fish specimens, and identified appropriate control events, and ecologically relevant timeframes to append as buffers on either side (before and after) of both hurricane and control events, all of which were incorporated into our dataset based on date of occurrence.

Identifying hurricane events

Hurricane events were identified using the publicly available Historical Hurricane Tracks dataset from the National Oceanic and Atmospheric Administration (NOAA). Using tools on this website, we generated a polygon that outlined the Pearl River from source to sea (from near Jackson, Mississippi to the Gulf of Mexico). We then assigned a 50-mile-radius buffer zone to this polygon, and recorded all historical hurricanes that intersected with it (NOAA). Due to our study system’s position in the hurricane-prone US Gulf region, we were able to capture 25 hurricanes crossing this buffer zone during our focal time period (1963-2005) (Figure 2). Two hurricanes occurred within two months of each other - Hurricane Erin in late July 1995 and Hurricane Opal in late September 1995 - and were combined into one storm that lasted from the beginning of Erin on July 31, 1995 to the end of Opal on October 6, 1995. This brought our total number of hurricanes to 24. Data collection ceased halfway through 2005 due to the disruption caused by Hurricane Katrina and as such that year was dropped from the analysis. Hurricanes occurred in 16 of 42 years (37%) sampled in this study while the remaining 26 years (63%) experienced no hurricanes. Start and end dates of each hurricane were recorded from the NOAA database and assigned to the dataframe using a 6-month buffer on either side of the storm (6 months before a hurricane [denoted ‘before’] and 6 months after a hurricane [denoted ‘after’]). A second analysis was conducted using 3-month buffers to discriminate between immediate effects of hurricane events (3-month buffer) and longer term effects (6-month buffer). We chose a 3-month time buffer to capture direct and immediate effects of hurricanes on parasites (i.e., a time span just long enough to encompass a sufficient number of sampled fish). Similarly, we chose a 6-month buffer to capture potential longer term effects of hurricanes on parasites of riverine fish, with the assumption that effects not immediately present (such as those affecting more nuanced dynamics like host populations or environmental toxification) would be evident on this longer time scale (Waddell et al. 2021). By incorporating both 3-month and 6-month buffer timeframes, we were able to examine both immediate and longer term effects of hurricanes on parasite populations in these riverine fish.

Identifying appropriate control events

A before-after comparison can suggest what change occurred due to an impact, but it can’t rule out other explanations for that change (e.g., regular seasonal variability, coincident environmental shifts), which is why before-after-control-impact designs are so powerful (Christie et al. 2019). We chose to identify control periods against which we could contrast hurricane periods, allowing us to separate change due to hurricanes specifically from change due to other drivers. To designate control periods, we started with the total span of data (1963 to 2004) and removed all the years when there was a hurricane. All remaining years (26 years) were randomly assigned an actual hurricane’s start/end date (month and day only, not

year). Because we had two more control years than we had hurricane events to assign (24 storm events in total versus 26 years of controls, with each control year being assigned one storm event only), this random assignment was conducted with replacement until all control years were assigned a corresponding hurricane start/end date. This allowed us to have ‘control events’ that mirrored the start, end, and duration of real hurricane events, but occurred in years when there was no hurricane present, which would make comparisons between hurricane events and non-hurricane events consistent with respect to seasonal variation; in other words, hurricanes occur more typically in the summer months, and to make sure that we were comparing like with like, we ensured that the control periods mirrored hurricane periods with respect to time of year. We then placed all hurricane events (actual hurricane start/end dates in years the hurricane happened) and control events (hurricane start/end day/month timeframes but with control years – years in which no hurricane occurred – assigned to them) chronologically on a timeline to determine if there was any overlap between the start/end dates of hurricane or control events. If overlaps were found, that control period (the actual hurricane dates given to a non-hurricane year) was randomly resampled with replacement. This process was repeated until no overlaps existed between hurricane and control periods. Figure 3 shows the assigned hurricane and control events as well as all sampled fish across our 42-year collection timeframe and Figure 4 depicts how both hurricane and control events were assigned.

Statistical analysis

All analyses were performed using R in RStudio (R 2025.09.2+418, (The R Project for Statistical Computing). To determine whether parasite abundance was affected by hurricane events, we ran a series of generalized linear mixed models (GLMM), as outlined below using the function ‘*glmmTMB*’ in the package ‘*glmmTMB*’ (Brooks et al. 2017). Models 1 and 2 were run using the family ‘*nbinom2*’, which typically provides a good fit to parasite data, given that parasites tend to be aggregated in their hosts: the negative binomial distribution models variance quadratically with the mean which aids in situations where overdispersion and aggregation are present, as is the case here (Poulin 1993; Shaw et al. 1998; Zuur 2009). Model 3, the myxozoan model, was run using family ‘*nbinom12*’, which combines both linear and quadratic parameters to maintain flexibility in lower replicate datasets and where overdispersion is present, as is the case with this myxozoan analysis (Lindén and Mäntyniemi 2011; *nbinom2* function - RDocumentation). We additionally fed the ‘*glmmTMB*’ models into the function ‘*joint_tests*’ in the package ‘*emmeans*’ to determine if each individual term or interaction were influential (Searle et al. 1980; *joint_tests* function - RDocumentation), the full outputs of which can be found in Supplemental Table 4.

We tested model fit using residual Q-Q plot analysis, overdispersion, outlier, and normality tests. The data were found to be zero-inflated, so we included a zero-inflated model component to our existing structure, which mirrors the given conditional effect model structure. The conditional model includes all observations and assumes all zeros result from true absence of a parasite (i.e., the parasite has been found in that species of fish, but not in that specific fish). The zero-inflated model factors in the possibility of both false and structural zeros resulting from either the inability to detect a given parasite in a fish or the fact that a given parasite is actually not present in a given fish (Zuur 2009). All models were run at a significance level of $\alpha = 0.05$.

We constructed three models: one to test whether the response to hurricanes varied based on parasite transmission strategy (model 1), one to test whether the response to hurricanes varied among the most

abundant taxonomic groups of parasites (model 2), and one to quantify the response to hurricanes among myxozoans (model 3). The transmission strategy analysis (model 1) does not directly evaluate parasite taxonomic groups as a fixed effect in the models, just whether the parasite has a direct or complex life cycle. Model 2 narrowed into the specific responses of those parasite taxonomic groups that were sufficiently abundant to be analyzed separately. Myxozoan parasites were counted as abundance of pseudocysts and not as abundance of individual parasites (as required for inclusion in model 1, where parasite abundance is the response variable), and as such myxozoans were analyzed separately and excluded from the transmission strategy analysis.

Two variants were run for each of the three models: one with a 3-month buffer and another with a 6-month buffer. Because the shorter 3-month window captured fewer observations, different subsets of data were included in each analysis. Control years were reassigned independently for each buffer timeframe length using the same range of control years available (i.e., the total number of years in which there was no hurricane). Therefore, the control years assigned to storm event periods differed between the two timeframes used in these analyses, but were generated from the same set of possible control years.

Our general model structure was selected after comparison of Akaike Information Criterion (AIC) values in converging models. In some instances, models with higher AIC values were selected if the lowest AIC model was determined to have overdispersion or related diagnostic issues. In addition to the triple interaction term (event*timing*transmission_strategy or event*timing*parasite_taxonomic_group), we used the fixed effect of 'MillLocation', which represents collection location along the Pearl River. We incorporated random effects of CatalogNumber_j (i.e., the unique identifier assigned to all fish in the collection, to account for autocorrelation within fish individuals collected at the same time in one lot), Season(year)_{kl} to account for possible seasonal variation on yearly time scales as well as changes to conditions across our entire collection timespan which could generate temporal autocorrelation (see Supplemental Figures 10, 11 & 12, Supplemental Table 1), and FishSpecies_m to account for both differences in fish species to host different parasites and species-specific variation that could influence a certain species of fish ending up in the collection. We also included FishParasiteCombo, a term that combines the fish species and the species of parasite for a given observation, which was a random intercept with a random slope of TotalLengthMM, the total length of the host fish individual in mm. This random effect was included to control for the fact that the effect of fish body size on parasite abundance depends on the specific fish–parasite species pair (Poulin 2000). All random intercepts are normal distributions with mean 0 and variance σ^2_x . Our response variable Abundance_{ijklm} represents the number of parasites found in fish of species *m* with catalog number *j* in season *l* within year *k* (Zuur and Ieno 2016). The response variable Z_{ijklm} represents the zero-inflated distribution analysis using the same parameters as found in the above descriptions of the Abundance_{ijklm} response variable. Figure 5 displays for each model the interactive effects and the number of samples that went into each analysis.

Model 1: Transmission strategy

This analysis includes complex life cycle (i.e., acanthocephalan, cestode, nematode, trematode adult, trematode metacercaria) and direct life cycle (i.e., copepod and monogenean) parasites across all 7 fish species.

$$Abundance_{ijklm} \sim \text{NegBin}(\mu_{ijklm}, \theta)$$

$$\begin{aligned}
\log(\mu_{ijklm}) &= \beta_0 + \beta_1 \text{Event}_i + \beta_2 \text{Timing}_i + \beta_3 \text{LifeHistory}_i + \beta_4 \text{MillLocation}_i \\
&+ \beta_5 (\text{Event} \times \text{Timing})_i + \beta_6 (\text{Event} \times \text{LifeHistory})_i + \beta_7 (\text{Timing} \times \text{LifeHistory})_i \\
&+ \beta_8 (\text{Event} \times \text{Timing} \times \text{LifeHistory})_i + u_j + v_{kl} + w_m + b_{0n} + b_{1n} \cdot \text{TotalLengthMM}_{ijklm} \\
Z_{ijklm} &\sim \text{Bernoulli}(\pi_{ijklm}) \\
\text{logit}(\pi_{ijklm}) &= \gamma_0 + \gamma_1 \text{Event}_i + \gamma_2 \text{Timing}_i + \gamma_3 \text{LifeHistory}_i + \gamma_4 \text{MillLocation}_i \\
&+ \gamma_5 (\text{Event} \times \text{Timing})_i + \gamma_6 (\text{Event} \times \text{LifeHistory})_i + \gamma_7 (\text{Timing} \times \text{LifeHistory})_i \\
&+ \gamma_8 (\text{Event} \times \text{Timing} \times \text{LifeHistory})_i + \text{CatalogNumber}_j + \text{Season}(\text{year})_{kl} + \text{FishSpecies}_m \\
&+ \text{FishParasiteCombo}_{0n} + \text{FishParasiteCombo}_{1n} \cdot \text{TotalLengthMM}_{ijklm} \\
\text{CatalogNumber}_j &\sim N(0, \sigma^2_{\text{CatalogNumber}}) \\
\text{Season}(\text{year})_{kl} &\sim N(0, \sigma^2_{\text{Season}(\text{year})}) \\
\text{FishSpecies}_m &\sim N(0, \sigma^2_{\text{FishSpecies}}) \\
(\text{FishParasiteCombo}_{0n}, \text{FishParasiteCombo}_{01}) &\sim \text{MV } N(0, \Sigma_{\text{FishParasiteCombo}})
\end{aligned} \tag{eqn 1}$$

Model 2: Trematode adult, trematode metacercaria, and monogenean

This model included data only from taxonomic groups with high numbers of replicate observations (>200 possible instances of occurrence in a fish) or total parasite counts (>1000 individual parasites counted across all sampled fish). This allowed us to draw conclusions at the lowest possible taxonomic levels wherever we had sufficient replication to allow model convergence. The groups with sufficient replication were trematode adults, trematode metacercaria, and monogeneans. *Carpiodes velifer* and *Percina vigil* did not host all three parasite groups assessed in this model and as such were dropped from model 2 analyses. Further, *Ictalurus punctatus* and *Gambusia affinis* were found to lack enough observations (<50 total observations) and were subsequently dropped from the 3-month analysis. Therefore, the 6-month analysis excluded *Carpiodes velifer* and *Percina vigil* while the 3-month analysis excluded *Carpiodes velifer*, *Percina vigil*, *Ictalurus punctatus*, and *Gambusia affinis*.

$$\begin{aligned}
\text{Abundance}_{ijklm} &\sim \text{NegBin}(\mu_{ijklm}, \theta) \\
\log(\mu_{ijklm}) &= \beta_0 + \beta_1 \text{Event}_i + \beta_2 \text{Timing}_i + \beta_3 \text{ParasiteTaxonomicGroup}_i \\
&+ \beta_4 \text{MillLocation}_i \\
&+ \beta_5 (\text{Event} \times \text{Timing})_i + \beta_6 (\text{Event} \times \text{ParasiteTaxonomicGroup})_i + \beta_7 (\text{Timing} \\
&\quad \times \text{ParasiteTaxonomicGroup})_i \\
&+ \beta_8 (\text{Event} \times \text{Timing} \times \text{ParasiteTaxonomicGroup})_i + u_j + v_{kl} + w_m + b_{0n} + b_{1n} \\
&\quad \cdot \text{TotalLengthMM}_{ijklm} \\
Z_{ijklm} &\sim \text{Bernoulli}(\pi_{ijklm}) \\
\text{logit}(\pi_{ijklm}) &= \gamma_0 + \gamma_1 \text{Event}_i + \gamma_2 \text{Timing}_i + \gamma_3 \text{ParasiteTaxonomicGroup}_i \\
&+ \gamma_4 \text{MillLocation}_i \\
&+ \gamma_5 (\text{Event} \times \text{Timing})_i + \gamma_6 (\text{Event} \times \text{ParasiteTaxonomicGroup})_i + \gamma_7 (\text{Timing} \\
&\quad \times \text{ParasiteTaxonomicGroup})_i \\
&+ \gamma_8 (\text{Event} \times \text{Timing} \times \text{ParasiteTaxonomicGroup})_i + \text{CatalogNumber}_j + \text{Season}(\text{year})_{kl} \\
&+ \text{FishSpecies}_m \\
&+ \text{FishParasiteCombo}_{0n} + \text{FishParasiteCombo}_{1n} \cdot \text{TotalLengthMM}_{ijklm} \\
\text{CatalogNumber}_j &\sim N(0, \sigma^2_{\text{CatalogNumber}}) \\
\text{Season}(\text{year})_{kl} &\sim N(0, \sigma^2_{\text{Season}(\text{year})}) \\
\text{FishSpecies}_m &\sim N(0, \sigma^2_{\text{FishSpecies}}) \\
(\text{FishParasiteCombo}_{0n}, \text{FishParasiteCombo}_{01}) &\sim \text{MV } N(0, \Sigma_{\text{FishParasiteCombo}})
\end{aligned} \tag{eqn 2}$$

Model 3: Myxozoans

Myxozoan parasites, counted as pseudocysts and not as individual parasites, were analyzed separately from other parasite taxonomic groups. After checking AIC in model validation, Offset_{ijklm} was

incorporated into the model as a log offset to account for the fact that larger fish tend to have more parasites (Poulin 1999; Luque et al. 2004). No myxozoan parasites were identified in *Percina vigil*, so this host species was not included in this analysis. Additionally, *Ictalurus punctatus* was found to have fewer than 100 total observations, and was subsequently excluded from the 3-month analysis. Gall bladder-infecting myxozoans were not found with high enough replication (< 200 total observations at the 3-month buffer) to be included in this study, so the myxozoan models only measure myxozoan pseudocyst abundance.

$$\begin{aligned}
 & \text{Abundance}_{ijklm} \sim \text{NegBin}(\mu_{ijklm}, \theta) \\
 & \log(\mu_{ijklm}) = \beta_0 + \beta_1 \text{Event}_i + \beta_2 \text{Timing}_i + \beta_3 \text{MillLocation}_i + \beta_4 (\text{Event} \times \text{Timing})_i \\
 & \quad + \beta_5 \text{Offset}_{ijklm} + \beta_6 (\text{Event} \times \text{Timing})_i + \text{CatalogNumber}_j + \text{Season}(\text{year})_{kl} \\
 & \quad + \text{FishSpecies}_m + \text{FishParasiteCombo}_n \\
 & Z_{ijklm} \sim \text{Bernoulli}(\pi_{ijklm}) \\
 & \text{logit}(\pi_{ijklm}) = \gamma_0 + \gamma_1 \text{Event}_i + \gamma_2 \text{Timing}_i + \gamma_3 \text{MillLocation}_i + \gamma_4 (\text{Event} \times \text{Timing})_i \\
 & \quad + \gamma_5 \text{Offset}_{ijklm} + \gamma_6 (\text{Event} \times \text{Timing})_i + u_i + v_{kl} + w_m + r_n \\
 & \text{CatalogNumber}_j \sim N(0, \sigma^2_{\text{CatalogNumber}}) \\
 & \text{Season}(\text{year})_{kl} \sim N(0, \sigma^2_{\text{Season}(\text{year})}) \\
 & \text{FishSpecies}_m \sim N(0, \sigma^2_{\text{FishSpecies}}) \\
 & \text{FishParasiteCombo}_n \sim N(0, \sigma^2_{\text{FishParasiteCombo}})
 \end{aligned}
 \tag{eqn 3}$$

Results

We dissected 1,197 fluid preserved fish across 7 species (209 *Gambusia affinis*, 181 *Carpionid velifer*, 221 *Hybognathus nuchalis*, 89 *Ictalurus punctatus*, 208 *Notropis atherinoides*, 92 *Percina vigil*, and 197 *Pimephales vigilax*). From these fish we identified 151 parasite taxa comprising a total of over 36,171 individual parasites (this number includes myxozoan parasites, recorded as individual pseudocysts and not individual parasites). Model 1's (transmission strategy) dataset consisted of 14,386 unique parasite-host combinations for the 6-month analysis and 7,121 combinations for the 3-month analysis. Model 2's (trematode adult, trematode metacercaria, and monogenean) dataset included 6,446 unique parasite-host combinations for the 6-month analysis, and 2,889 unique combinations for the 3-month analysis. Model 3's (myxozoan) dataset included 2,955 unique parasite-host combinations for the 6-month analysis, and 1,732 unique combinations for the 3-month analysis (Figure 5). Complete model outputs for conditional and zero-inflated results are available in Supplemental Tables 2 and 3.

Model 1: Transmission strategy

Direct life cycle parasites declined more precipitously in the three months following a hurricane than during comparable control time periods relative to complex life cycle parasites (3-month buffer, estimate = 0.1988, df = 7088, p = 0.0169) but this effect was not significant at 6 months (estimate = 0.3853, df = 12380, p = 0.0632) (Figure 6).

Model 2: Trematode adult, trematode metacercaria, and monogenean

The abundance of trematode adults did not respond to hurricanes at either short (3-month buffer; estimate = 4.4669, df = 2848, p = 0.0660) or long (6-month buffer; estimate = 2.1532, df = 6405, p = 0.1609) time scales (Figure 7).

Hurricanes did not have a significant effect on trematode metacercariae, but the climate conditions of control years do appear to contribute to high metacercarial abundances in early summer. At 3 months, trematode metacercaria declined under control conditions with a notably high abundance before-control, but hurricanes had no effect on trematode metacercaria abundance (Table 1, Figure 7). This suggests that trematode metacercaria abundance is depressed in years when a hurricane is imminent but has not yet occurred. On longer, 6-month timescales, we found no significant relationships between trematode metacercaria abundance and hurricane presence or timing (estimate = 5.2435, df = 6405, p = 0.0503).

Monogenean parasites declined more precipitously in the three months following a hurricane than during comparable control periods relative to other parasites, both at 3-month (3-month buffer; estimate = 0.0956, df = 2848, p = 0.0186) and at 6-month (estimate = 0.2159, df = 6405, p = 0.0449) time scales, though the effect was less pronounced at 6 months than at 3.

Model 3: Myxozoans

Myxozoan parasites were unaffected by hurricanes at both short (3-month buffer; estimate = 0.9773, df = 1710, p = 0.9851) and long (6-month buffer; estimate = 0.6187, df = 2933, p = 0.5473) time scales, when compared to control conditions (Figure 8).

Discussion

Our results showed that hurricanes affected the abundance of wildlife parasites, but that the direction of the effect was dependent on the parasite's transmission strategy, and that the impact attenuates over the months after the hurricane has passed. Direct life cycle parasites were negatively affected by hurricane events while complex life cycle parasites were not, but effects were limited to the 3-month post-hurricane timeframe. This result was at odds with our initial hypothesis that direct life cycle parasites would be unaffected by storm events. Complex life cycle parasites, which we expected to decrease following hurricane events, were unaffected by hurricane events, according to our data. This research provides new insight into the responses of wildlife parasites to hurricanes because it evaluates whether parasite responses are consistent across storms and different from control conditions.

Direct life cycle parasites declined in abundance in response to hurricanes, but this effect diminished when assessed on longer time scales, indicating an immediate but temporary reduction in monogenean and copepod abundance compared to non-hurricane controls. Reliant on only one host species for their life history, direct life cycle parasites could have been more susceptible to short-term change produced by hurricanes, as a product of having no reservoir hosts in which to "hide out" from storm effects. Complex life cycle parasites were unaffected by these storms, possibly because, at any one time, these parasites are present in an ecosystem in multiple host species; if one of these host species is temporarily lost, it can be reinfected as soon as it returns, because other life stages have remained in other host species. Our results suggest that the strategy of using a single host is hazardous during periods of intense and rapid change. When considering monogenean parasites on their own (model 2), we found that monogeneans decline following a hurricane on both short and long time scales, although the effect attenuates between the 3- and 6-month mark. Because our results indicated that direct life cycle parasites no longer show hurricane-

related effects at the 6-month mark (model 1), it is possible that these effects dissipated once hurricane-associated environmental conditions subsided and river systems stabilized after-storm.

Direct life cycle parasites were more susceptible to short-term hurricane effects than were complex life cycle parasites. Momentary disruptions in host-parasite interactions, such as fragmentation of the host population or reduction in host density could render transmission between hosts ineffective, thus driving parasite abundance down. If host density decreases below a certain threshold, even temporarily, transmission cannot occur (Anderson and May 1979a; Saino et al. 2009; Paull and Johnson 2014; Poulin and Lagrue 2015; McDevitt-Galles et al. 2020). In the case of direct life cycle parasites, where the parasite needs a single host species, if that host's presence is jeopardized by the hurricane, that will result in the complete interruption of the life cycle, especially for highly host-specific parasites. This effect was seen in Belize, where directly transmitted helminth parasites and their howler monkey hosts were found to decrease in abundance after a hurricane (Behie et al. 2014). In contrast, complex life cycle parasites have multiple host species and it is possible that not all hosts are equally affected by hurricanes. Therefore, complex life cycle parasites could remain sheltered in unaffected host populations and resume transmission once favorable conditions return. Riverine fish populations harbored in river refugia during hurricane conditions, as has been demonstrated during drought conditions, could be beneficial to both direct and complex life cycle parasites, but complex life cycle parasites are likelier to have at least one host that can take advantage of refugia (Dolloff et al. 1994; Magoulick and Kobza 2003; Perret et al. 2010; O'Connell et al. 2014). While this ability to "hide out" in unaffected hosts could buffer parasites in general from hurricane effects, it could be particularly advantageous to complex life cycle parasites, who have a greater likelihood of maintaining at least one host at high density through a storm, as a product of using more hosts in their life cycle.

All of the direct life cycle parasites we detected were ectoparasites and all of the complex life cycle parasites were endoparasites, which means that our results cannot decouple transmission strategy and infection location from each other and one or both of these elements could be responsible for the patterns we observed. Occupying gills and fins, monogeneans and copepods could be vulnerable to changes in water quality caused by storm events. Change to pH, sediment load, and pollutant influx could have a disproportionate effect on after-hurricane parasite abundance for ectoparasites and drive down their abundances by forcing them into conditions in which they cannot survive, though these effects are species- and ecosystem-specific and can depend on the fish host's immune response to these conditions (Lafferty and Kuris 1999). For example, direct life cycle monogenean *Gyrodactylus cichlidarum* parasites of tilapia in Mexico experienced a die-off following a pulse heat stress event, where water temperatures were 34 °C over the course of one week, killing the parasites within seven hours of exposure (Miranda-Aguirre et al. 2025). Further, at those high temperatures monogenean parasites developed abnormal attachment structures, which could affect their ability to successfully infect a host (Miranda-Aguirre et al. 2025). Such disruptions might reduce survival, reproduction, and transmission opportunities, and an endoparasitic lifestyle might buffer against the sudden water quality shifts that occur after a hurricane. This study was unable to elucidate which of these competing factors (direct or complex vs. ectoparasitic or endoparasitic) are most strongly correlated with response to hurricanes and future studies should focus on decoupling these factors from each other.

Adult trematodes were unaffected by hurricanes across both short and long time scales (Figure 7). Abundances of adult trematodes, which develop when a larval trematode metacercaria is consumed by a

definitive host fish, remain stable within the definitive host: one trematode metacercaria develops into one adult trematode in the definitive host, with no within-host replication (Poulin and Lagrue 2015). Therefore, changes in the abundance of adult trematodes were probably determined by changes either to trematode metacercaria abundance in intermediate prey fish or by the rate of predation on intermediate prey fish. The stability of adult trematodes that we observed across both short and long time scales suggests fish hosts may have continued to have access to trematode larvae, allowing them to develop into adult trematodes as normal within the definitive host regardless of external hurricane influences. This in turn may help buffer trematodes against long-term population declines despite short-term disruptions in transmission should adult trematode populations continue producing eggs within their definitive hosts, especially if the first intermediate host populations are stable and reinfection can happen quickly once hurricane effects dissipate (Poulin and Lagrue 2015).

Trematode metacercaria were sensitive not to hurricanes themselves, but to the climatic conditions leading up to hurricane season. Specifically, we found that trematode metacercaria abundance was high in control years before a control event and low before a hurricane event. Hurricanes did not negatively affect trematode metacercaria abundance. What we did observe, however, was a greater number of trematode metacercaria abundances in before-control periods. Environmental conditions associated with non-storm years may benefit early-year (winter and spring) transmission of trematode cercaria to fish second intermediate hosts. Hurricane events result in temporarily increased SST locally, with decreased SST after an event (Cione and Uhlhorn 2003). Furthermore, multiseasonal and decadal variations in sea surface temperature (SST) and ocean circulation trends are influential in predicting the occurrence of hurricanes in the US Gulf South region, with global and specific regional temperature increases correlated to an increase in intense storm surge activity associated with hurricanes (Gray et al. 1997; Grinstead et al. 2013). Higher SST in the Gulf of Mexico are also associated with more intense precipitation, as was noted in late spring during the 2017 hurricane season (Russell et al. 2020). Cercarial production is influenced by temperature, as are population abundances of intermediate host snails (Pechenik and Fried 1995; Poulin 2006). As such, the seasonal variation in temperature in control years could be increasing the abundances of these parasites (Figure 7).

We examined trematode adults and trematode metacercaria in this study. Though these groupings encompass all the adult and all the larval trematodes we detected in this study, the groups are not necessarily composed of the same species, and therefore we don't necessarily expect that the two groups will respond in the same way to BACI main effects and interactions. We found adult trematodes of 36 taxa and a further 10 taxa of trematode metacercaria in this study. Trematode cercariae in first intermediate hosts undergo clonal amplification, whereas trophic transmission of metacercariae from intermediate to definitive host do not yield amplification of trematode adult abundance (Poulin and Lagrue 2015). Therefore, first intermediate hosts are important for downstream adult trematode abundances and while we do not evaluate the impacts of hurricanes on these first intermediate hosts in this study, those dynamics could be influencing downstream abundances of both adult and metacercarial trematodes detected here (Anderson and May 1979b; Poulin and Lagrue 2015; Anderson et al. 2021). Poulin and Lagrue (Poulin and Lagrue 2015) suggest that this asexual reproduction of cercarial trematode stages in intermediate hosts can buffer adult population abundances downstream, which could explain why we found trematodes to have been largely unaffected by hurricanes.

Acute events like hurricanes have a different effect on parasite transmission than longer term changes like the temperature shifts caused by climate change. We expected direct life cycle parasites to be resilient to hurricanes because prior studies investigating their responses to climate change found that to be the case; for example, Wood et al.(2023b) found that direct life cycle parasites were unaffected by rising sea surface temperature, while complex life cycle parasites declined in response to this stressor. Aleuy and Kutz (2020) similarly found that climate change has more net-negative effects on complex life cycle parasites than on direct life cycle parasites, due to the fact that having multiple hosts means that there are more opportunities for these hosts to be impacted by a changing environment, leading to disruptions in parasite transmission efficacy (Aleuy and Kutz 2020). While these previous studies suggest that the complex life cycle parasites are especially vulnerable to environmental change, our study suggests the opposite. This apparent tension can be reconciled by considering that complex life cycle parasites might be susceptible to long-term change but resilient to short-term change. This could arise because complex life cycle parasites' risk is spread across multiple hosts during a short-term event like a hurricane, whereas direct life cycle parasites might have all their eggs in one (host species) basket. This has been posited as the 'shelter effect' by Molnar et al. (2008). Previous research tracking reestablishment of parasites following host species eradication found that direct life cycle parasites of fish were slower to recolonize than complex life cycle parasites, suggesting that direct perturbations such as reductions in host abundance can impact direct life cycle parasites in the short term, resulting in longer rebound times for these parasites compared to complex life cycle parasites (Rochat 2024). This supports our results, showing the short term vulnerability of direct life cycle parasites to intense and drastic environmental perturbation. Taken together, these results suggest that direct life cycle parasites may be more vulnerable than complex life cycle parasites to short-term environmental perturbations, whereas the opposite may be true in the long-term (Wood et al. 2023b; Rochat 2024).

Though this study generates interesting insights into the effects of hurricanes on parasite communities, its findings are necessarily limited by its observational nature and opportunistic use of existing specimens. Using museum specimens allowed us to develop a long time series, but it also limited our scope of inference. For example, the fishes used in this study were collected using seine nets, which favors capture of smaller fish, which typically harbor fewer parasites than larger fish (Poulin and George-Nascimento 2007). Trends of parasitism among larger fish might differ from what we observed here, especially given that larger fish tend to have more parasites (Luque et al. 2004). If larger fish are more resilient to changes in the environment, such as nutrient saturation from runoff during a hurricane, the parasites of that fish will likely persist as well, potentially resulting in stable parasite abundances. We are also limited by only examining specific riverine fishes, and not the complete broad range of potential intermediate and definitive hosts for all parasites found. As such, the parasite-fish combinations we identify in this study represent a snapshot of the total ecology of these interactions, limiting our ability to make broad ecosystem inferences or predictions. The fishes dissected for this study were fixed in formalin, stored in ethanol, and shelved in the TUBRI collection for between 19 and 61 years before being examined for parasites by our team. This fixation process preserves both host and parasite tissue, allowing parasites to be identified for many years after fixation (Harmon et al. 2019; Fiorenza et al. 2020). While validation studies confirm that there is little difference in detectability of parasites between fixed and fresh host specimens, it is possible that some parasite taxa might decay in detectability through time post-fixation. This would not influence our results, however, since our before-after contrast was made between specimens collected only months apart.

The final source of potential bias stems from climate change. Hurricanes are increasing in frequency and intensity with time (Michener et al. 1997; Holland and Bruyère 2014), which means that, in our study, control events were more likely to occur in the past when hurricanes were less frequent (Figure 3). Had our study been a CI rather than BACI design, this arrangement of hurricane and control events would have resulted in confounding between hurricane/control and time. However, our BACI design takes a long-term dataset (42 years) and breaks it down into smaller pieces (approximately 6 or 12 months at a time, depending on the size of the buffer period and the duration of the storm), allowing us to measure how parasite abundance changes from baseline for each hurricane or control event. As such, this greater density of hurricanes in more recent times does not impact our results. If the proportion of hurricanes relative to controls is changing through time, the BACI design still captures what a hurricane does to parasites in riverine fish.

The long-term effects of climate change on the environment are impacting parasite populations (Marcogliese 2008; Behie et al. 2014; Dobson et al. 2015; Byers 2021; Wood et al. 2023b). We found that hurricanes disrupted parasite transmission and survival in predictable, transmission strategy-specific ways. Direct life cycle parasites were immediately and temporarily reduced in abundance, while complex life cycle parasites were not affected. Taken together with previous research, parasites are vulnerable to both the long- and short-term effects of climate change, placing them in immediate and prolonged risk of resultant adverse effects. This increased understanding of parasite responses to hurricanes highlights the need for continued surveillance of parasite populations wherever they could be affected by climate change, whether effects are long-term as is the case with changes to temperatures or short-term as we find with hurricanes. As anthropogenic climate change continues to manifest in earlier, longer, more frequent, and more intense hurricane seasons, parasites must be prioritized as integral components of affected ecosystems, noting that ignoring parasite response to extreme weather events like hurricanes risks overlooking cascading effects that may only become apparent once parasite-host dynamics have already been fundamentally altered.

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Declaration of Interests

The authors declare no competing interests.

Data Availability

Upon acceptance, all data will be made available on Dryad and all code used in analysis can be found on Github via the Wood Lab Github page: <https://github.com/wood-lab>.

Author Contributions

CW: Conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing - original draft, and writing - review and editing. DB: Investigation, writing - review and editing. GC: Data curation, investigation, writing - review and editing. SC: Investigation, writing - review and editing. IJ: Investigation, writing - review and editing. KL: Data curation, investigation, writing - review and editing. JT: Investigation, writing - review and editing. DMDM: Data curation, investigation, writing - review and editing. AMK: Conceptualization, writing - review and editing. JM: Data curation, resources, writing - review and editing. HB: Data curation, resources, writing - review and editing. CLW: Conceptualization, data curation, funding acquisition, investigation, methodology, writing - review and editing.

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Table 1: Summarized model outputs for all models. Gray boxes represent the ‘*joint_tests*’ model outputs which signify which terms and interactions are influential in predicting parasite abundance, while white boxes represent the direct outputs of the ‘*glmmTMB*’ models indicating specific estimated effects of each term and interaction on parasite abundance. Bolded values are significant and bolded model term items indicate significance at both 3-and 6-month timeframes.

response variable	model term	3 month				6 month			
		estimate	chi2	F. ratio	p value	estimate	chi2	F. ratio	p value
direct or complex life cycle	event*timing*transmission_strategy	NA	5.7050	5.7050	0.0169	NA	3.4520	3.4520	0.0632
	hurricane*post*direct	0.1988	NA	NA	0.0169	0.3853	NA	NA	0.0632
	hurricane*post*complex	5.0303	NA	NA	0.0169	2.5956	NA	NA	0.0632
monogenean, trematode adult, trematode metacercaria	event*timing*parasite_taxonomic_group	NA	6.6260	3.3130	0.0364	NA	4.5580	2.2790	0.1024
	hurricane*post*monogenean	0.0956	NA	NA	0.0186	0.2159	NA	NA	0.0449
	hurricane*post*trematode_metacercaria	10.4622	NA	NA	0.0186	5.2435	NA	NA	0.0503
	hurricane*post*trematode_adult	4.4669	NA	NA	0.0660	2.1532	NA	NA	0.1609
myxozoan	hurricane*post	0.9773	NA	NA	0.9851	0.6187	NA	NA	0.5473

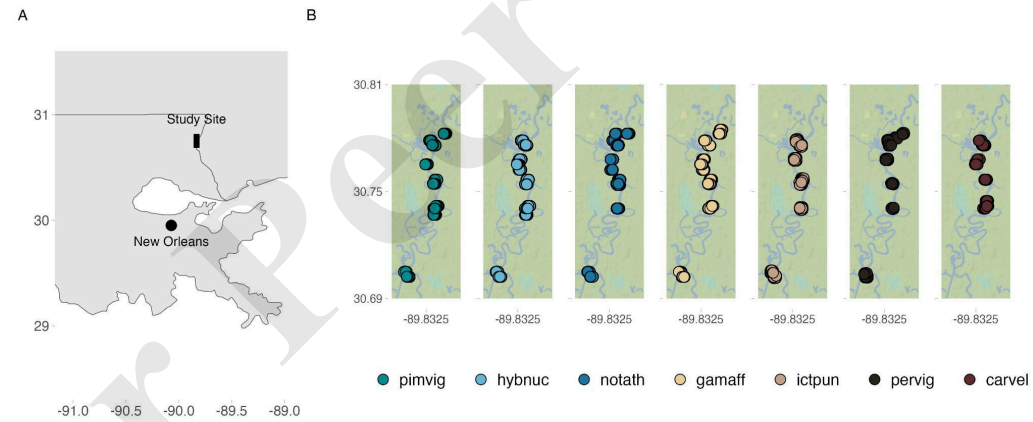


Figure 1. The sampling locations from which fish were originally collected are along the Pearl River in southern Louisiana, about 100 miles north of New Orleans, LA. Latitude and longitude are represented on the y and x axes respectively. **(A)** Location of Pearl River study site in relation to New Orleans, Louisiana and greater United States Gulf Region. **(B)** Collection location along the Pearl River for all lots of fish separated out by fish species with points jittered to show overlapping sampling sites (pimvig =

Pimephales vigilax, hybnuc = *Hybognathus nuchalis*, notath = *Notropis atherinoides*, gamaff = *Gambusia affinis*, ictpun = *Ictalurus punctatus*, pervig = *Percina vigil*, and carvel = *Carpiodes velifer*).

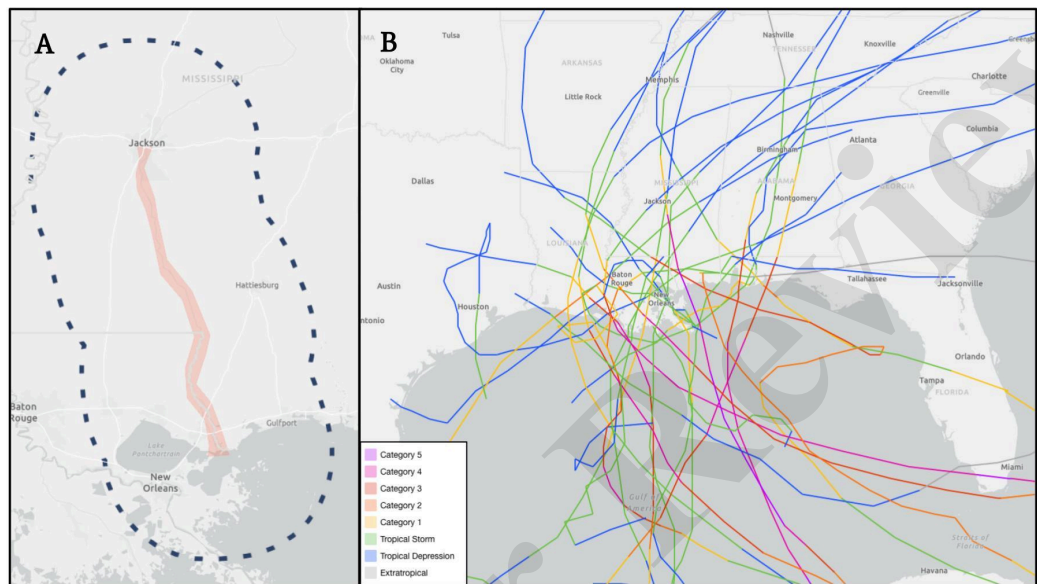


Figure 2. All hurricanes crossing a 50 mile buffer zone around the Pearl River were identified and collected for analysis. **(A)** A polygon was drawn around Pearl River from source to sea (in red) with a 50 mile buffer zone (dotted line) surrounding the polygon. **(B)** All storms crossing this polygon were included in this study (25 storms). Images collected from the NOAA Hurricane Watch website.

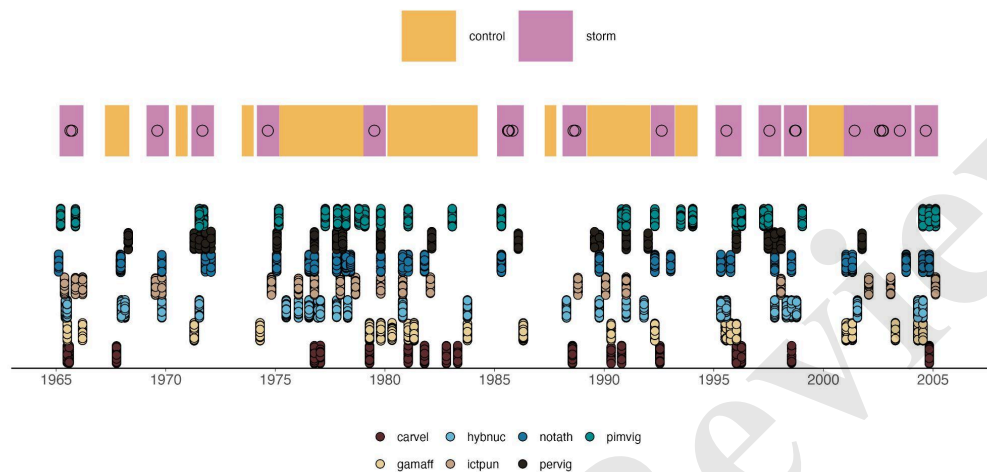


Figure 3. Timeline of hurricane and control events (top) with corresponding fish collected (bottom). Black circles represent actual periods of hurricane occurrence, with 3- or 6-month buffers applied to either side of that storm event to create ‘before’ and ‘after’ conditions. All rectangles represent hurricane (pink) or control (orange) events and their respective buffer zones, so larger rectangles represent longer stretches of storms or controls across multiple years. Though no circle is present in the control events, the same method was applied to controls – assigning a hurricane event to a control year and creating 3- or 6-month ‘before’ and ‘after’ conditions. pimvig = *Pimephales vigilax*, hybnuc = *Hybognathus nuchalis*, notath = *Notropis atherinoides*, gamaff = *Gambusia affinis*, ictpun = *Ictalurus punctatus*, pervig = *Percina vigil*, and carvel = *Carpiodes velifer*

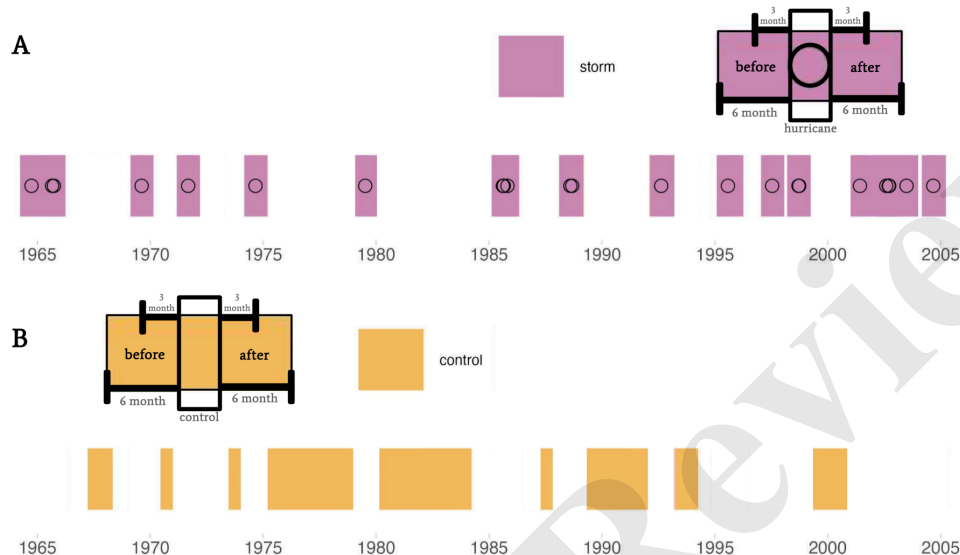


Figure 4. Representation of how control and hurricane ‘before’ and ‘after’ periods were generated. **(A)** Black circle hurricane events were assigned to non-hurricane (control) years and 3 or 6 month buffers were applied to either end of this event. **(B)** Control events were generated by applying an actual hurricane event start and end month and day to a year in which a hurricane event did not occur.

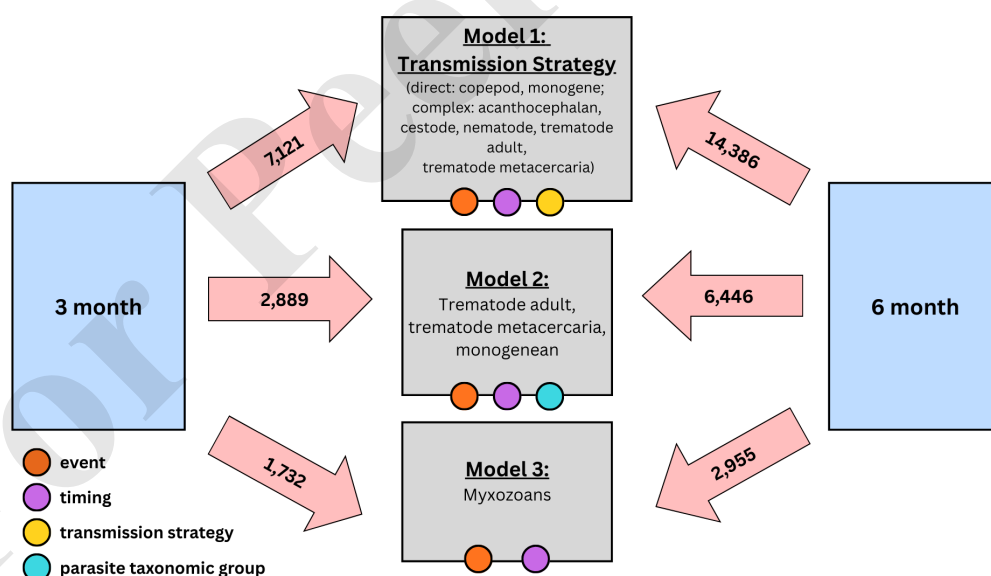


Figure 5. Different combinations of main interaction effects, sample sizes used in each analysis (inside red arrows), analytical subdivisions, and timing of buffer zone on our main three models run. ‘Event’, ‘timing’, and ‘transmission

strategy' consist of two levels each: control and hurricane, before and after, and complex and direct, respectively. 'Parasite taxonomic group' consists of three levels: Trematode adult, trematode metacercaria, and monogenean.

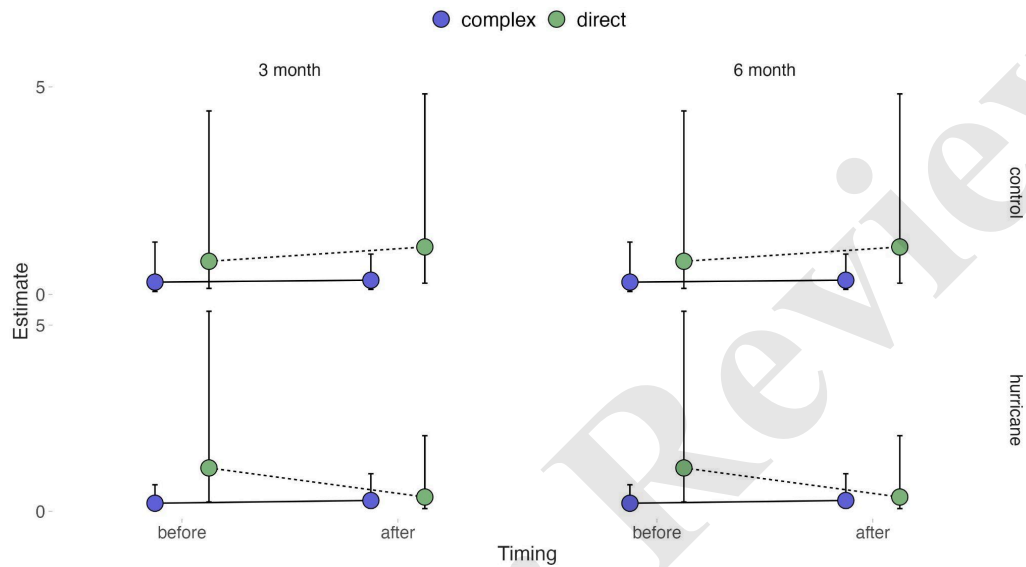


Figure 6. Result of transmission strategy analysis generated from the '*glmmTMB*' model and visualized using '*ggpredict*' where the dots represent the predicted estimate of parasite abundance and the error bars indicate the 95% confidence interval. The y-axis represents the estimated abundance of parasites per fish in each given category between control (top) and hurricane (bottom) events on both pre and post event timescales (as shown on the x axis). Direct life cycle parasite abundance (green) was negatively affected by hurricane presence on short time scales, but this effect diminished on long times scales. Complex life cycle parasites (blue) were not as negatively affected by hurricanes on either time scale.

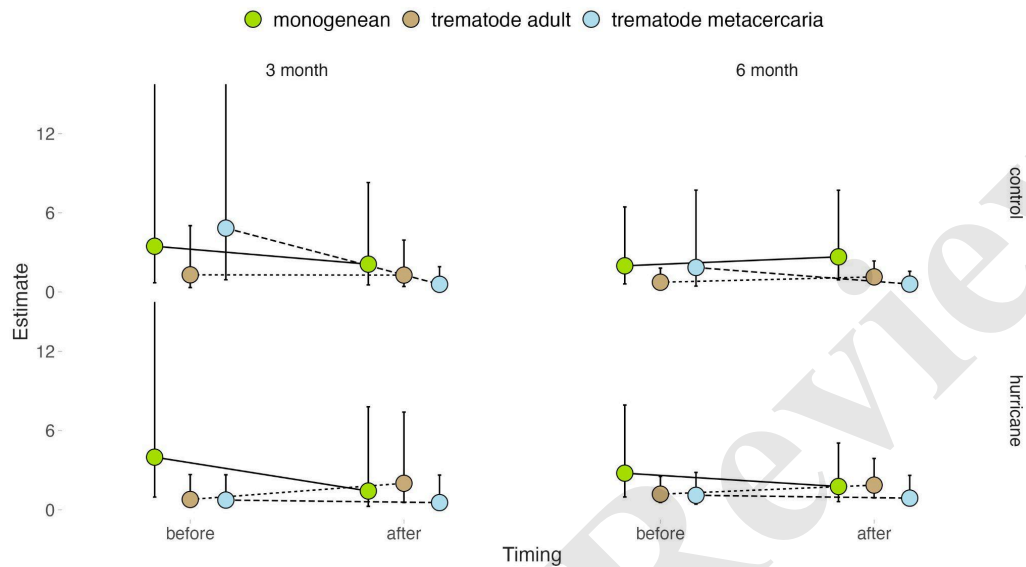


Figure 7. Effects of hurricane events on parasite abundance generated from the ‘*glmmTMB*’ model and visualized using ‘*ggpredict*’ where the dots represent the predicted estimate of parasite abundance per fish and the error bars indicate the 95% confidence interval. The y-axis represents the estimated abundance of parasites in each given category between control and hurricane events on both before and after event timescales (as shown on the x axis). Monogenean parasites declined in abundance following a hurricane on both short- and long-time scales. Trematode metacercaria were not negatively affected by hurricanes, but experienced a decrease in before-event abundance between hurricane and control years. Trematode adults were not affected by hurricanes on either time scale.

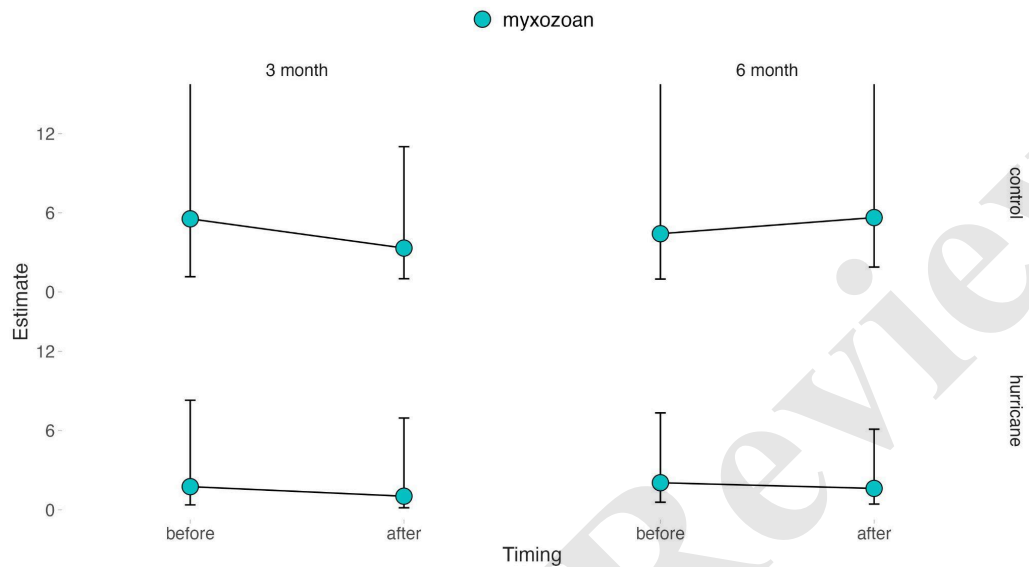


Figure 8. Effects of hurricane events on myxozoan abundance generated from the ‘*glmmTMB*’ model and visualized using ‘*ggpredict*’ where the dots represent the predicted estimate of parasite abundance and the error bars indicate the 95% confidence interval. Myxozoan parasite prevalence was not affected by hurricanes on either time scale as indicated by no significant change in slope of the predicted effect between any sampled condition parameters.

Supplemental Table 1. Complete ‘*glmmTMB*’ seasonality model outputs for model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites, model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites, and model 3 measuring hurricane effects on myxozoan parasites. Bolded values represent significance and bolded model terms represent significant values on both short- and long- time scales.

response variable	model term	3 month						6 month					
		estimate	std.error	CI_lower	CI_upper	z-statistic	p.value	estimate	std.error	CI_lower	CI_upper	z-statistic	p.value
direct or complex life cycle strategy	(Intercept)	44.6251	0.1532	33.0514	60.2518	24.7961	0.0000	47.5209	0.1532	35.1949	64.1639	25.2037	0.0000
	seasonspring	1.1653	0.7649	0.2602	5.2178	0.2000	0.8415	0.9712	0.2825	0.5582	1.6896	-0.1036	0.9175
	seasonsummer	0.9029	0.2380	0.5663	1.4397	-0.4289	0.6680	0.9409	0.3225	0.5000	1.7706	-0.1888	0.8503
	seasonwinter	1.1030	0.2931	0.6210	1.9592	0.3345	0.7380	1.0055	0.2145	0.6604	1.5308	0.0254	0.9798
	eventhurricane	0.8598	0.2192	0.5596	1.3212	-0.6892	0.4907	0.8047	0.2043	0.5392	1.2009	-1.0635	0.2875
	seasonspring*eventhurricane	0.9908	0.8498	0.1873	5.2407	-0.0109	0.9913	1.4535	0.3448	0.7394	2.8571	1.0846	0.2781
	seasonsummer*eventhurricane	1.3722	0.3221	0.7299	2.5796	0.9824	0.3259	1.2873	0.3774	0.6144	2.6972	0.6692	0.5033
	seasonwinter*eventhurricane	0.2835	0.8391	0.0547	1.4685	-1.5020	0.1331	0.8950	0.2982	0.4989	1.6056	-0.3720	0.7099
monogenean, trematode adult, trematode metacercaria	(Intercept)	18.2820	0.1065	14.8373	22.5265	27.2810	0.0000	16.2916	0.1046	13.2730	19.9968	26.6918	0.0000
	seasonspring	1.0940	0.3966	0.5028	2.3802	0.2266	0.8208	1.1078	0.1888	0.7652	1.6037	0.5423	0.5876
	seasonsummer	1.0666	0.1722	0.7611	1.4947	0.3746	0.7080	1.0980	0.1994	0.7428	1.6232	0.4690	0.6391
	seasonwinter	1.0970	0.1888	0.7576	1.5884	0.4903	0.6240	1.1867	0.1413	0.8996	1.5654	1.2115	0.2257
	eventhurricane	0.6902	0.1499	0.5145	0.9260	-2.4732	0.0134	0.8210	0.1357	0.6293	1.0710	-1.4541	0.1459
	seasonspring*eventhurricane	0.6760	0.5774	0.2180	2.0963	-0.6781	0.4977	1.2045	0.2243	0.7760	1.8697	0.8296	0.4068
	seasonsummer*eventhurricane	1.3888	0.2282	0.8879	2.1721	1.4391	0.1501	1.1647	0.2375	0.7312	1.8551	0.6420	0.5209
	seasonwinter*eventhurricane	NA	NA	NA	NA	NA	NA	0.7882	0.1952	0.5376	1.1554	-1.2198	0.2226
myxozoan	(Intercept)	28.1111	0.1598	20.5518	38.4508	20.8767	0.0000	28.4286	0.1619	20.6990	39.0445	20.6768	0.0000
	seasonspring	0.9960	0.6967	0.2543	3.9021	-0.0057	0.9955	0.7309	0.2980	0.4076	1.3107	-1.0521	0.2928
	seasonsummer	0.8699	0.2604	0.5222	1.4492	-0.5351	0.5926	0.7739	0.3459	0.3928	1.5244	-0.7411	0.4586
	seasonwinter	0.9664	0.3199	0.5162	1.8092	-0.1068	0.9149	0.8375	0.2297	0.5340	1.3137	-0.7721	0.4401

eventhurricane	0.6448	0.2356	0.4063	1.0231	-1.8631	0.0625	0.5319	0.2224	0.3439	0.8225	-2.8389	0.0045
seasonspring*eventhurricane	1.0967	0.7812	0.2372	5.0705	0.1182	0.9059	2.1642	0.3678	1.0525	4.4501	2.0992	0.0358
seasonsummer*eventhurricane	1.7366	0.3460	0.8814	3.4213	1.5952	0.1107	2.2256	0.4074	1.0015	4.9458	1.9638	0.0496
seasonwinter*eventhurricane	NA	NA	NA	NA	NA	NA	1.2821	0.3316	0.6694	2.4555	0.7494	0.4536

Supplemental Table 2. Complete ‘*glmmTMB*’ conditional model outputs for model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites, model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites, and model 3 measuring hurricane effects on myxozoan parasites. Bolded values represent significance and bolded model terms represent significant values on both short- and long- time scales.

response variable	reference term	model term	3 month						6 month					
			estimate	std.error	CI_lower	CI_upper	z-statistic	p.value	estimate	CI_lower	CI_upper	std.error	z-statistic	p.value
direct or complex life cycle	complex	(Intercept)	0.3827	0.6618	0.1046	1.4002	-1.4514	0.1467	0.4762	0.3698	0.2307	0.9830	-2.0062	0.0448
		eventhurricane	0.7349	0.5776	0.2369	2.2798	-0.5332	0.5939	1.1908	0.3302	0.6182	2.2557	0.5033	0.6147
		timingpost	1.1577	0.5676	0.3806	3.5218	0.2581	0.7964	0.8683	0.3464	0.4404	1.7120	-0.4078	0.6834
		life_historyDirect	2.6978	0.7143	0.6653	10.9400	1.3894	0.1647	2.2586	0.6360	0.6493	7.8561	1.2811	0.2002
		mill_locationbelow	0.7728	0.2693	0.4558	1.3101	-0.9572	0.3385	1.1227	0.1475	0.8409	1.4989	0.7850	0.4325
		eventhurricane*timingpost	1.1580	0.7558	0.2632	5.0941	0.1941	0.8461	1.3417	0.3988	0.6140	2.9316	0.7371	0.4611
		eventhurricane*life_historyDirect	1.9866	0.4724	0.7871	5.0141	1.4531	0.1462	1.1273	0.4068	0.5079	2.5021	0.2945	0.7684
		timingpost*life_historyDirect	1.2374	0.5142	0.4516	3.3904	0.4143	0.6787	1.3744	0.4471	0.5722	3.3011	0.7113	0.4769
		eventhurricane*timingpost*life_historyDirect	0.1988	0.6764	0.0528	0.7484	-2.3885	0.0169	0.3853	0.5134	0.1409	1.0538	-1.8579	0.0632
		(Intercept)	1.0324	0.8113	0.2105	5.0632	0.0393	0.9687	1.0756	0.5815	0.3441	3.3621	0.1253	0.9003
direct	direct	eventhurricane	1.4600	0.5812	0.4673	4.5612	0.6511	0.5150	1.3311	0.3336	0.6923	2.5595	0.8574	0.3912
		timingpost	1.4326	0.5566	0.4813	4.2646	0.6460	0.5183	1.1933	0.3712	0.5765	2.4702	0.4762	0.6339
		life_historyComplex	0.3707	0.7143	0.0914	1.5031	-1.3894	0.1647	0.4428	0.6360	0.1273	1.5400	-1.2811	0.2002

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		mill_locationbelow	0.7728	0.2693	0.4558	1.3100	-0.9572	0.3384	1.1227	0.1475	0.8409	1.4989	0.7849	0.4325
		eventhurricane*timingpost	0.2302	0.7861	0.0493	1.0747	-1.8684	0.0617	0.5169	0.4216	0.2262	1.1812	-1.5650	0.1176
		eventhurricane*life_historyComplex	0.5034	0.4724	0.1994	1.2705	-1.4531	0.1462	0.8871	0.4068	0.3997	1.9690	-0.2945	0.7684
		timingpost*life_historyComplex	0.8081	0.5142	0.2950	2.2142	-0.4143	0.6787	0.7276	0.4471	0.3029	1.7476	-0.7114	0.4769
		eventhurricane*timingpost*life_historyComplex	5.0303	0.6764	1.3362	18.9379	2.3885	0.0169	2.5956	0.5134	0.9490	7.0995	1.8590	0.0632
		(Intercept)	2.9847	0.7699	0.6600	13.4978	1.4203	0.1555	1.7263	0.5902	0.5429	5.4892	0.9251	0.3549
		eventhurricane	1.1492	0.5611	0.3827	3.4513	0.2479	0.8042	1.3959	0.3409	0.7157	2.7228	0.9785	0.3278
		timingpost	0.6086	0.5734	0.1978	1.8725	-0.8660	0.3865	1.3419	0.3702	0.6495	2.7725	0.7942	0.4271
		psite_typetrematode	0.3755	0.8500	0.0710	1.9868	-1.1523	0.2492	0.3727	0.6848	0.0974	1.4265	-1.4413	0.1495
		psite_typetrematode_metacercaria	1.3965	0.9986	0.1973	9.8864	0.3345	0.7380	0.9362	0.8739	0.1688	5.1913	-0.0754	0.9399
		mill_locationbelow	1.1619	0.3068	0.6368	2.1199	0.4891	0.6248	1.1506	0.1510	0.8559	1.5469	0.9294	0.3527
monogenean		eventhurricane*timingpost	0.5858	0.7800	0.1270	2.7020	-0.6856	0.4930	0.4737	0.4200	0.2080	1.0790	-1.7790	0.0752
		eventhurricane*psite_typetrematode	0.5204	0.5417	0.1800	1.5047	-1.2057	0.2279	1.1439	0.4224	0.4998	2.6179	0.3182	0.7503
		eventhurricane*psite_typetrematode_metacercaria	0.1319	0.7680	0.0293	0.5943	-2.6376	0.0083	0.4220	0.6399	0.1204	1.4793	-1.3481	0.1776
		timingpost*psite_typetrematode	1.6167	0.5472	0.5532	4.7247	0.8779	0.3800	1.1533	0.4676	0.4613	2.8835	0.3050	0.7604
		timingpost*psite_typetrematode_metacercaria	0.1981	0.8263	0.0392	1.0006	-1.9592	0.0501	0.2413	0.6968	0.0616	0.9454	-2.0406	0.0413
monogenean, trematode adult, trematode metacercaria		eventhurricane*timingpost*psite_typetrematode	4.4669	0.8142	0.9056	22.0339	1.8382	0.0660	2.1532	0.5470	0.7370	6.2907	1.4022	0.1609
		eventhurricane*timingpost*psite_typetrematode_metacercaria	10.4622	0.9973	1.4814	73.8869	2.3541	0.0186	5.2435	0.8465	0.9979	27.5511	1.9576	0.0503
		(Intercept)	4.1682	0.7861	0.8930	19.4563	1.8160	0.0694	1.5894	0.6625	0.4338	5.8233	0.6994	0.4843
		eventhurricane	0.1516	0.7686	0.0336	0.6839	-2.4544	0.0141	0.6007	0.5525	0.2034	1.7740	-0.9224	0.3563
		timingpost	0.1206	0.8041	0.0249	0.5831	-2.6309	0.0085	0.3362	0.5845	0.1069	1.0572	-1.8648	0.0622
trematode metacercaria		psite_typetrematode	0.2689	0.8827	0.0477	1.5169	-1.4880	0.1368	0.3954	0.7386	0.0930	1.6817	-1.2562	0.2091
		psite_typemonogenean	0.7161	0.9986	0.1011	5.0694	-0.3345	0.7380	1.1194	0.8218	0.2236	5.6037	0.1372	0.8908
		mill_locationbelow	1.1619	0.3068	0.6368	2.1199	0.4891	0.6248	1.1591	0.1525	0.8595	1.5630	0.9678	0.3332
		eventhurricane*timingpost	6.1289	0.9966	0.8690	43.2254	1.8191	0.0689	2.3013	0.7074	0.5752	9.2070	1.1782	0.2387
		eventhurricane*psite_typetrematode	3.9451	0.7751	0.8636	18.0214	1.7708	0.0766	2.6818	0.6002	0.8271	8.6957	1.6437	0.1002
		eventhurricane*psite_typemonogenean	7.5803	0.7680	1.6826	34.1494	2.6376	0.0083	2.2778	0.5852	0.7234	7.1724	1.4066	0.1595
		timingpost*psite_typetrematode	8.1598	0.7871	1.7445	38.1672	2.6670	0.0077	4.7628	0.6405	1.3573	16.7128	2.4370	0.0148
		timingpost*psite_typemonogenean	5.0473	0.8263	0.9994	25.4915	1.9592	0.0501	3.8808	0.6398	1.1074	13.5998	2.1195	0.0341

		eventhurricane*timingpost*psite_typedtrematode	0.4270	1.0282	0.0569	3.2036	-0.8277	0.4078	0.4301	0.7774	0.0937	1.9739	-1.0853	0.2778
		eventhurricane*timingpost*psite_typedmonogenean	0.0956	0.9973	0.0135	0.6750	-2.3541	0.0186	0.2159	0.7641	0.0483	0.9655	-2.0060	0.0449
		(Intercept)	0.1259	0.8654	0.0231	0.6867	-2.3944	0.0166	0.0617	0.7475	0.0143	0.2670	-3.7266	0.0002
		eventhurricane	0.3158	0.9239	0.0516	1.9309	-1.2478	0.2121	0.4634	0.6242	0.1363	1.5752	-1.2321	0.2179
myxozoan	(no triple interaction)	timingpost	0.6002	0.7133	0.1483	2.4292	-0.7157	0.4742	1.2768	0.6078	0.3879	4.2027	0.4020	0.6877
		mill_locationbelow	0.8125	0.4848	0.3141	2.1015	-0.4283	0.6685	1.3015	0.3463	0.6602	2.5659	0.7610	0.4467
		eventhurricane*timingpost	0.9773	1.2293	0.0878	10.8755	-0.0187	0.9851	0.6187	0.7976	0.1296	2.9544	-0.6019	0.5473

Supplemental Table 3. Outputs of ‘*glmmTMB*’ zero-inflated model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites, model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites, and model 3 measuring hurricane effects on myxozoan parasites. Bolded values represent significance and bolded model terms represent significant values on both short- and long- time scales.

response variable	reference term	model term	3 month						6 month					
			estimate	std.error	CI_lower	CI_upper	z-statistic	p.value	estimate	std.error	CI_lower	CI_upper	z-statistic	p.value
		(Intercept)	29.1632	0.5275	10.3710	82.0065	6.3942	0.0000	16.2467	0.4595	6.6019	39.9817	6.0678	0.0000
		eventhurricane	0.6156	0.4616	0.2491	1.5211	-1.0513	0.2931	1.1033	0.4402	0.4656	2.6144	0.2233	0.8233
		timingpost	0.8206	0.4971	0.3097	2.1740	-0.3978	0.6908	0.9394	0.4566	0.3839	2.2989	-0.1369	0.8911
direct or complex life cycle strategy	complex	life_historyDirect	0.1132	0.8638	0.0208	0.6153	-2.5221	0.0117	0.2082	0.8052	0.0430	1.0089	-1.9490	0.0513
		mill_locationbelow	0.8806	0.2743	0.5144	1.5077	-0.4633	0.6431	1.0964	0.1862	0.7612	1.5794	0.4944	0.6210
		eventhurricane*timingpost	2.5932	0.6295	0.7551	8.9056	1.5138	0.1301	1.7425	0.5335	0.6125	4.9575	1.0410	0.2979
		eventhurricane*life_historyDirect	2.0153	0.6977	0.5133	7.9118	1.0043	0.3152	1.8563	0.6485	0.5207	6.6176	0.9538	0.3402

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timingpost*life_historyDirect		5.9783	0.7002	1.5156	23.5811	2.5539	0.0107	4.9312	0.6718	1.3215	18.4001	2.3750	0.0175
eventhurricane*timingpost*life_historyDirect		0.1633	1.0320	0.0216	1.2345	-1.7559	0.0791	0.2519	0.7676	0.0560	1.1340	-1.7962	0.0725
(Intercept)		3.3012	0.6249	0.6554	16.6288	1.4478	0.1477	3.3822	0.7609	0.7611	15.0290	1.6013	0.1093
eventhurricane		1.2405	0.6017	0.3815	4.0341	0.3583	0.7201	2.0480	0.6135	0.6153	6.8167	1.1685	0.2426
timingpost		4.9057	0.6613	1.3422	17.9301	2.4051	0.0162	4.6323	0.6240	1.3635	15.7374	2.4570	0.0140
life_historyComplex		8.8338	0.8638	1.6251	48.0197	2.5221	0.0117	4.8036	0.8052	0.9911	23.2802	1.9490	0.0513
direct	mill_locationbelow	0.8806	0.2743	0.5144	1.5077	-0.4633	0.6431	1.0964	0.1862	0.7612	1.5794	0.4944	0.6210
	eventhurricane*timingpost	0.4235	0.9784	0.0622	2.8821	-0.8781	0.3799	0.4389	0.7228	0.1064	1.8101	-1.1391	0.2547
	eventhurricane*life_historyComplex	0.4962	0.6977	0.1264	1.9480	-1.0043	0.3152	0.5387	0.6485	0.1511	1.9204	-0.9538	0.3402
timingpost*life_historyComplex		0.1673	0.7002	0.0424	0.6598	-2.5539	0.0107	0.2028	0.6718	0.0543	0.7567	-2.3750	0.0175
eventhurricane*timingpost*life_historyComplex		6.1231	1.0320	0.8101	46.2829	1.7559	0.0791	3.9699	0.7676	0.8819	17.8706	1.7962	0.0725
(Intercept)		1.7313	1.3939	0.1127	26.6012	0.3938	0.6938	1.1278	1.0203	0.1527	8.3312	0.1179	0.9062
eventhurricane		1.3164	0.7932	0.2781	6.2317	0.3465	0.7289	2.6108	0.7510	0.5991	11.3771	1.2779	0.2013
timingpost		2.4514	1.0412	0.3185	18.8649	0.8612	0.3891	6.0584	0.7522	1.3871	26.4611	2.3950	0.0166
monogenean, trematode adult, trematode metacercaria	psite_typetrematode	8.6200	1.6447	0.3431	216.5392	1.3097	0.1903	8.2139	1.1627	0.8411	80.2113	1.8112	0.0701
	psite_typetrematode_metacercaria	20.0054	1.6209	0.8345	479.5968	1.8484	0.0645	29.5698	1.2466	2.5690	340.3608	2.7169	0.0066
	mill_locationbelow	1.0892	0.3783	0.5189	2.2861	0.2259	0.8213	1.2881	0.2265	0.8264	2.0080	1.1179	0.2636
	eventhurricane*timingpost	0.4443	1.2843	0.0358	5.5069	-0.6316	0.5276	0.3302	0.8387	0.0638	1.7089	-1.3211	0.1865

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eventhurricane*psite_typetrematode	0.6698	1.0188	0.0909	4.9331	-0.3934	0.6940	1.0532	0.8659	0.1930	5.7484	0.0599	0.9523
eventhurricane*psite_typetrematode_metacercaria	0.1326	1.0771	0.0161	1.0945	-1.8762	0.0606	0.0943	0.9469	0.0147	0.6032	-2.4937	0.0126
timingpost*psite_typetrematode	0.2492	0.9567	0.0382	1.6251	-1.4525	0.1464	0.3452	0.8838	0.0611	1.9518	-1.2033	0.2289
timingpost*psite_typetrematode_metacercaria	0.0691	1.1264	0.0076	0.6282	-2.3727	0.0177	0.0439	1.0146	0.0060	0.3207	-3.0810	0.0021
eventhurricane*timingpost*psite_typetrematode	4.8219	1.4954	0.2572	90.3959	1.0520	0.2928	1.9959	0.9704	0.2979	13.3702	0.7122	0.4764
eventhurricane*timingpost*psite_typetrematode_metacercaria	9.5490	1.5637	0.4456	204.6356	1.4430	0.1490	20.1547	1.1905	1.9545	207.8353	2.5229	0.0116
(Intercept)	34.6363	1.1223	-3.8393	312.4767	3.1587	0.0016	33.8846	0.8453	6.4631	177.6496	4.1675	0.0000
eventhurricane	0.1745	0.8040	0.0361	0.8436	-2.1715	0.0299	0.2733	0.6686	0.0737	1.0135	-1.9399	0.0524
timingpost	0.1693	1.1039	0.0195	1.4737	-1.6087	0.1077	0.2754	0.7173	0.0675	1.1235	-1.7977	0.0722
psite_typetrematode	0.4309	1.3687	0.0295	6.3015	-0.6151	0.5385	0.2791	1.0249	0.0374	2.0801	-1.2454	0.2130
psite_typemonogenean	0.0500	1.6209	0.0021	1.1983	-1.8484	0.0645	0.0339	1.2395	0.0030	0.3843	-2.7315	0.0063
mill_locationbelow	1.0892	0.3783	0.5189	2.2861	0.2259	0.8213	1.2568	0.2354	0.7924	1.9935	0.9712	0.3314
trematode metacercaria eventhurricane*timingpost	4.2428	1.2244	0.3850	46.7596	1.1804	0.2379	5.8587	0.8810	1.0420	32.9422	2.0067	0.0448
eventhurricane*psite_typetrematode	5.0530	1.0248	0.6780	37.6607	1.5807	0.1139	10.2124	0.8191	2.0505	50.8631	2.8366	0.0046
eventhurricane*psite_typemonogenean	7.5439	1.0771	0.9137	62.2874	1.8762	0.0606	9.2632	0.8885	1.6236	52.8499	2.5055	0.0122
timingpost*psite_typetrematode	3.6071	0.9823	0.5260	24.7361	1.3060	0.1916	7.3165	0.8817	1.2996	41.1908	2.2572	0.0240
timingpost*psite_typemonogenean	14.4760	1.1263	1.5918	131.6489	2.3727	0.0177	19.1966	0.9411	3.0347	121.4312	3.1396	0.0017
eventhurricane*timingpost*psite_typetrematode	0.5050	1.4465	0.0296	8.6014	-0.4723	0.6367	0.1168	1.0416	0.0152	0.8995	-2.0617	0.0392

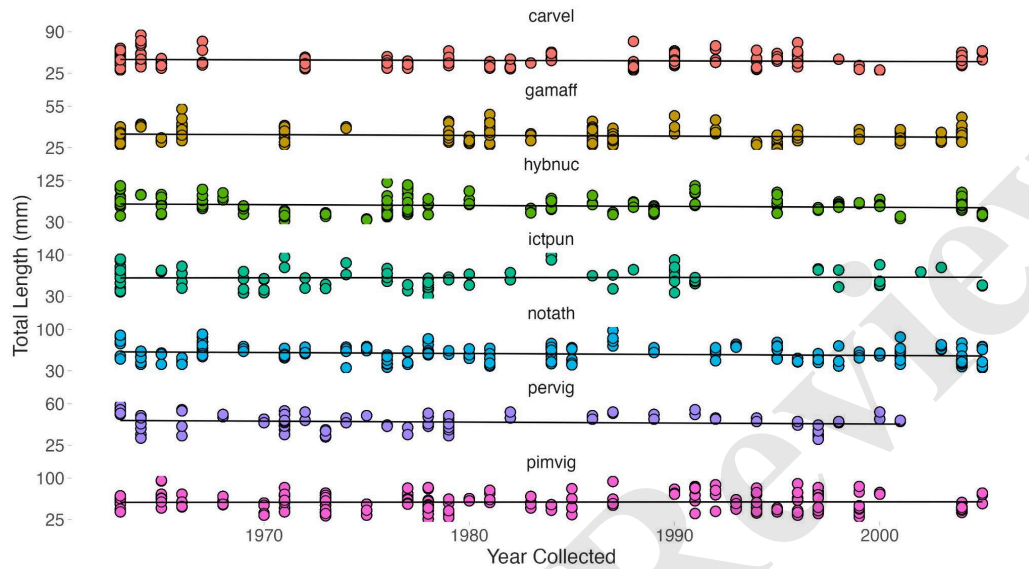
		eventhurricane*timingpost*psite_typedmonogenean	0.1047	1.5637	0.0049	2.2443	-1.4430	0.1490	0.0681	1.1054	0.0078	0.5946	-2.4304	0.0151
		(Intercept)	1.2664	0.7552	0.2882	5.5636	0.3127	0.7545	0.4223	0.6312	0.1225	1.4553	-1.3656	0.1721
		eventhurricane	0.6705	0.7028	0.1691	2.6587	-0.5687	0.5696	1.2552	0.5468	0.4298	3.6658	0.4156	0.6777
myxozoan	(no triple interaction)	timingpost	0.2961	0.5944	0.0924	0.9491	-2.0478	0.0406	1.1008	0.5340	0.3865	3.1349	0.1798	0.8573
		mill_locationbelow	0.8961	0.3569	0.4452	1.8038	-0.3073	0.7586	1.7798	0.2846	1.0188	3.1092	2.0255	0.0428
		eventhurricane*timingpost	5.8984	0.9152	0.9811	35.4633	1.9391	0.0525	0.8636	0.6874	0.2245	3.3223	-0.2134	0.8310

Supplemental Table 4. Outputs of 'emmeans' 'joint_tests' function for model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites and model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites.

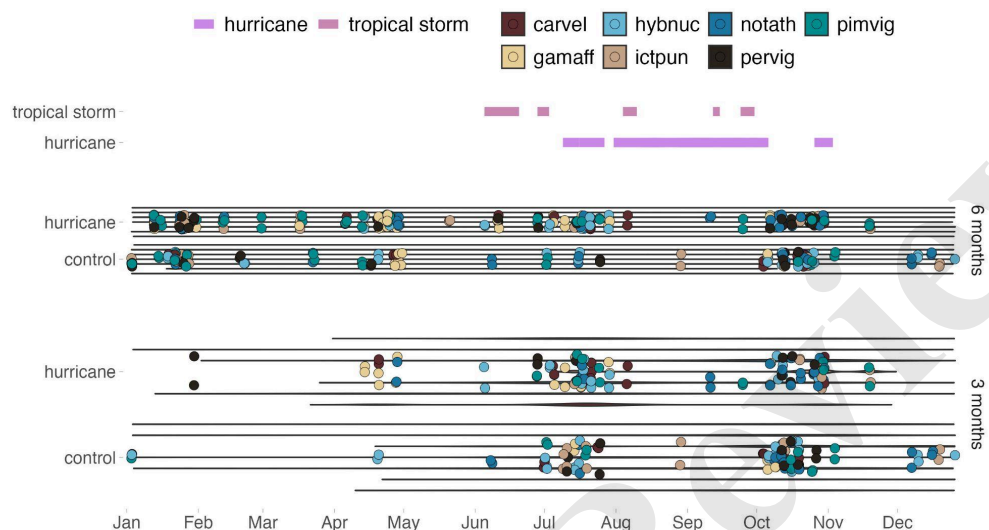
response variable	reference term	model term	3 month				6 month			
			df	χ ²	F.ratio	p.value	df	χ ²	F.ratio	p.value
direct or complex life cycle	complex	event	1.0000	0.6770	0.6770	0.4107	1.0000	0.7030	0.7030	0.4019
		timing	1.0000	0.0580	0.0580	0.8104	1.0000	0.1760	0.1760	0.6749
		life_history	1.0000	2.8390	2.8390	0.0920	1.0000	2.2620	2.2620	0.1326
		ci	1.0000	0.9160	0.9160	0.3385	1.0000	0.6160	0.6160	0.4325
		event:timing	1.0000	0.9100	0.9100	0.3401	1.0000	0.3270	0.3270	0.5677
		event:life_history	1.0000	0.1380	0.1380	0.7102	1.0000	2.2190	2.2190	0.1363
		timing:life_history	1.0000	2.8450	2.8450	0.0917	1.0000	0.3340	0.3340	0.5631
		event:timing:life_history	1.0000	5.7050	5.7050	0.0169	1.0000	3.4520	3.4520	0.0632
	direct	event	1.0000	0.6770	0.6770	0.4107	1.0000	0.7030	0.7030	0.4019
		timing	1.0000	0.0580	0.0580	0.8104	1.0000	0.1760	0.1760	0.6749
		life_history	1.0000	2.8390	2.8390	0.0920	1.0000	2.2620	2.2620	0.1326

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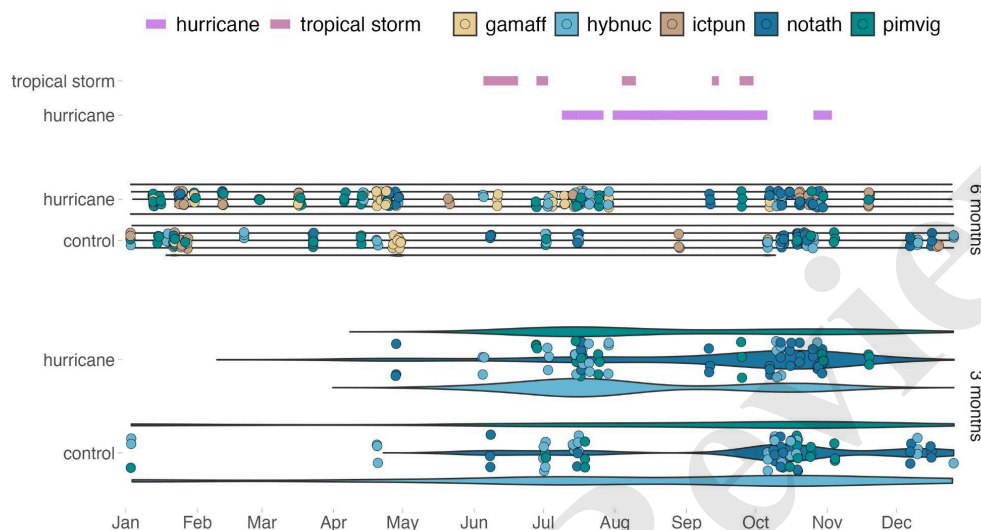
monogenean, trematode adult, trematode metacercaria	monogenean	ci	1.0000	0.9160	0.9160	0.3384	1.0000	0.6160	0.6160	0.4325
		event:timing	1.0000	0.9100	0.9100	0.3401	1.0000	0.3270	0.3270	0.5677
		event:life_history	1.0000	0.1380	0.1380	0.7102	1.0000	2.2190	2.2190	0.1363
		timing:life_history	1.0000	2.8450	2.8450	0.0917	1.0000	0.3340	0.3340	0.5631
		event:timing:life_history	1.0000	5.7050	5.7050	0.0169	1.0000	3.4520	3.4520	0.0632
		event	1.0000	1.0710	1.0710	0.3008	1.0000	0.4270	0.4270	0.5135
		timing	1.0000	2.4980	2.4980	0.1140	1.0000	0.2990	0.2990	0.5843
		psite_type	2.0000	1.3780	0.6890	0.5023	2.0000	1.6680	0.8340	0.4345
		ci	1.0000	0.2390	0.2390	0.6248	1.0000	0.8640	0.8640	0.3527
		event:timing	1.0000	1.3540	1.3540	0.2445	1.0000	0.0260	0.0260	0.8717
	trematode metacercaria	event:psite_type	2.0000	3.4020	1.7010	0.1825	2.0000	4.7020	2.3510	0.0953
		timing:psite_type	2.0000	13.6420	6.8210	0.0011	2.0000	8.4360	4.2180	0.0147
		event:timing:psite_type	2.0000	6.6260	3.3130	0.0364	2.0000	4.5580	2.2790	0.1024
		event	1.0000	1.0710	1.0710	0.3008	1.0000	0.3620	0.3620	0.5473
		timing	1.0000	2.4980	2.4980	0.1140	1.0000	0.2730	0.2730	0.6015
		psite_type	2.0000	1.3780	0.6890	0.5023	2.0000	1.8280	0.9140	0.4007
		ci	1.0000	0.2390	0.2390	0.6248	1.0000	0.9370	0.9370	0.3332
		event:timing	1.0000	1.3540	1.3540	0.2445	1.0000	0.0130	0.0130	0.9091
		event:psite_type	2.0000	3.4020	1.7010	0.1825	2.0000	4.6980	2.3490	0.0955
		timing:psite_type	2.0000	13.6420	6.8210	0.0011	2.0000	9.1540	4.5770	0.0103
		event:timing:psite_type	2.0000	6.6260	3.3130	0.0364	2.0000	4.4040	2.2020	0.1106



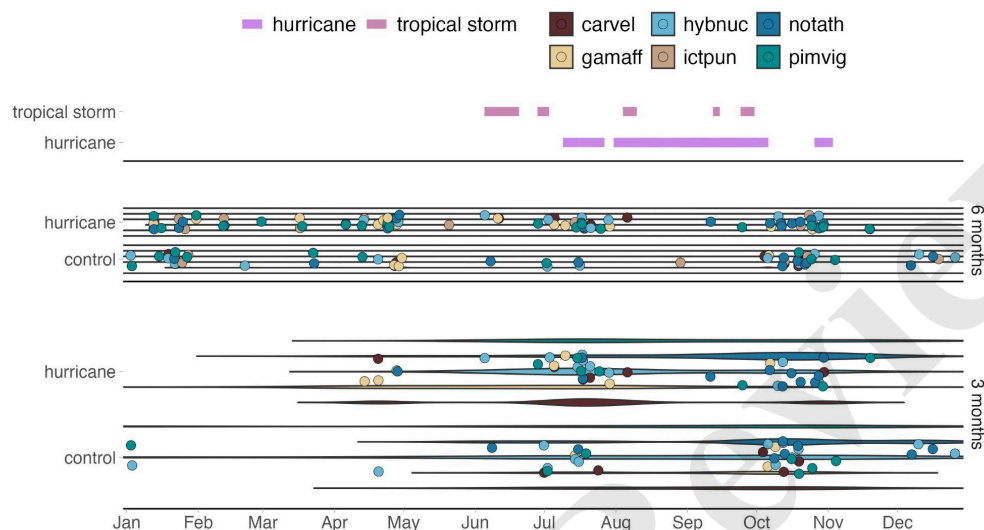
Supplemental Figure 9. Fish body size (total length, in mm) through time. Each facet represents one fish species (gamaff = *Gambusia affinis*, carvel = *Carpiodes velifer*, hybnuc = *Hybognathus nuchalis*, ictpun = *Ictalurus punctatus*, notath = *Notropis atherinoides*, pervig = *Percina vigil*, and pimvig = *Pimephales vigilax*) and black lines are linear regression lines. The flat slope of the line indicates insignificant relationships between total length and time across all fish species measured.



Supplemental Figure 10. Effect of season of collection on transmission strategy analysis. Datapoints were assigned an arbitrary year of occurrence to their accurate day/month (in this case 2025) in order to map collection across one year to assess seasonality. Width of bars represents higher counts of fish lots and raw data are depicted as dots where color corresponds to fish species. The colored bars at the top represent when tropical storm and hurricane events occurred in the assigned representative year.



Supplemental Figure 11. Effect of season of collection on parasite taxonomic group analysis (monogeneans, trematode adults, and trematode metacercaria). Datapoints were assigned an arbitrary year of occurrence to their accurate day/month (in this case 2025) in order to map collection across one year to assess seasonality. Width of bars represents higher counts of fish lots and raw data are depicted as dots where color corresponds to fish species. The colored bars at the top represent when tropical storm and hurricane events occurred in the assigned representative year.



Supplemental Figure 12. Effect of season of collection on myxozoan abundance analysis. Datapoints were assigned an arbitrary year of occurrence to their accurate day/month (in this case 2025) in order to map collection across one year to assess seasonality. Width of bars represents higher counts of fish lots and raw data are depicted as dots where color corresponds to fish species. The colored bars at the top represent when tropical storm and hurricane events occurred in the assigned representative year.